

# A Gauge Theory of Temporal Residuals

## Reference-Free Time, Observational Equivalence, and Temporal Curvature

Lord Soumadeep Ghosh and OpenAI

Kolkata, India

### Abstract

This paper develops a reference-free **Calculus of Temporal Residuals** (CTR) by importing the gauge-theoretic logic of observational equivalence into the analysis of distributed clock readings. A temporal observation is modeled as a scalar field  $\tau$  together with a temporal comparison one-form  $A$ . The familiar mean is treated as a projection and a convenient gauge fixing rather than as intrinsically privileged time. The invariant residual one-form is

$$\Theta = d\tau - A, \quad (1)$$

with curvature  $\Omega = d\Theta$ . Gauge orbits, quotient spaces, path-integrated residuals, holonomy, covariance kernels, spectral distortion, information divergence, and gauge-minimized temporal fakeness are defined. The framework connects residualization in statistics and econometrics with relativity, synchronization, complex-systems theory, finance, biology, psychology, welfare economics, political signaling, and machine learning. The paper is intentionally concise: standard theory is cited rather than rederived, while the new structure is stated with short proofs.

The paper ends with “The End”

## 1 Introduction

Suppose a clock is assigned to every event of a four-dimensional spacetime manifold  $M$ . Let

$$\tau : M \rightarrow \mathbb{R} \quad (2)$$

be the resulting temporal observation field. A conventional statistical analysis chooses a domain  $D \subseteq M$ , computes

$$\bar{\tau}_D = \frac{\int_D \tau d\mu}{\int_D d\mu}, \quad (3)$$

and decomposes

$$\tau = \bar{\tau}_D + \delta\tau_D. \quad (4)$$

The conventional story calls  $\bar{\tau}_D$  the common reference and often treats  $\delta\tau_D$  as error. Yet residuals can contain systematic structure, persistence, dependence, and information; econometrics makes residual diagnostics central, while modern statistical learning emphasizes representation and invariance [2–5].

A directly relevant precedent is Ghosh and OpenAI’s *Gauge-Invariant Measures of Fakeness under Logistic Regression*. That paper constructs nontrivial transformations that preserve logistic logits and decisions while materially changing coordinate representations, and interprets the model as operating on equivalence classes under a symmetry group [1]. Its conceptual lesson is that observational equivalence need not imply representational or ontological identity.

CTR applies that lesson to temporal observation. The central distinction is

$$\boxed{\text{temporal representation} \neq \text{reference-independent temporal structure.}} \quad (5)$$

The aim is not to claim that physical time is itself a fundamental gauge field. It is to construct a rigorous language in which temporal reference is explicit, admissible reference changes form a symmetry, and invariant temporal structure can be studied.

## 2 Reference-Dependent CTR

Define the averaging projection  $P_D$  and residual operator  $R_D$  by

$$P_D \tau = \frac{\int_D \tau \, d\mu}{\int_D d\mu}, \quad (6)$$

$$R_D = I - P_D. \quad (7)$$

Then

$$\delta\tau_D = R_D \tau = \tau - P_D \tau. \quad (8)$$

**Proposition 2.1** (Projection structure). *The reference and residual operators satisfy*

$$P_D^2 = P_D, \quad R_D^2 = R_D, \quad (9)$$

$$P_D R_D = 0, \quad R_D P_D = 0, \quad (10)$$

and

$$\int_D \delta\tau_D \, d\mu = 0. \quad (11)$$

*Proof.* Averaging a constant reproduces the constant, so  $P_D^2 = P_D$ . The other identities follow from  $R_D = I - P_D$ , and the zero-integral property follows immediately from the definition of  $P_D$ .  $\square$

The variance is merely a scalar compression of the residual field:

$$\sigma_{\tau,D}^2 = \frac{\int_D (\delta\tau_D)^2 \, d\mu}{\int_D d\mu}. \quad (12)$$

Thus the conceptual hierarchy is not mean  $\rightarrow$  error, but

$$\text{field} \rightarrow \text{projection} \rightarrow \text{residual field} \rightarrow \text{summaries}. \quad (13)$$

Table 1: Reference-dependent CTR objects.

| Object    | Definition              | Interpretation                        |
|-----------|-------------------------|---------------------------------------|
| Reference | $P_D \tau$              | Chosen common component               |
| Residual  | $R_D \tau$              | Mean-free temporal structure          |
| Gradient  | $\nabla \delta\tau_D$   | Local heterogeneity                   |
| Hessian   | $\nabla^2 \delta\tau_D$ | Local curvature of residual structure |
| Variance  | $\sigma_{\tau,D}^2$     | Scalar aggregate spread               |

### 3 Reference-Free CTR as a Gauge Theory

The problem with allowing an arbitrary residual reference is immediate: if the reference can be chosen as  $\tau$ , the residual becomes zero. A meaningful reference-free theory must therefore specify a group of admissible transformations and identify only those representations that differ by that group.

Let a temporal representation be

$$\mathcal{T} = (\tau, A), \quad (14)$$

where  $A$  is a one-form encoding the local rule for comparing temporal observations. Let  $\phi : M \rightarrow \mathbb{R}$  be smooth and define the reference transformation

$$\tau' = \tau + \phi, \quad (15)$$

$$A' = A + d\phi. \quad (16)$$

**Definition 3.1** (Gauge-invariant temporal residual). *The fundamental reference-free residual is*

$$\Theta = d\tau - A. \quad (17)$$

**Theorem 3.1** (Reference invariance). *For every admissible  $\phi$ ,*

$$\Theta' = \Theta. \quad (18)$$

*Proof.*

$$\Theta' = d(\tau + \phi) - (A + d\phi) \quad (19)$$

$$= d\tau - A \quad (20)$$

$$= \Theta. \quad (21)$$

□

The fundamental object is therefore not an absolute temporal value but a reference-invariant differential relation. Under a constant shift, this reduces to the familiar statement that temporal differences do not depend on the choice of time origin.

#### 3.1 Affine interpretation

A more geometric formulation places clock readings in an affine temporal bundle  $\mathcal{T}_{\text{aff}} \rightarrow M$ . An affine space has meaningful differences but no distinguished origin. A section gives a temporal representation; a comparison connection determines how values at different points are related. In a chosen trivialization, its covariant differential takes the form  $\Theta = d\tau - A$  [6–8]. This places the construction in the same broad mathematical family as gauge connections without asserting that  $A$  is a fundamental physical field.

### 4 Temporal Curvature and Holonomy

Define the temporal curvature two-form by

$$\Omega = d\Theta. \quad (22)$$

Because  $d^2\tau = 0$ ,

$$\Omega = -dA. \quad (23)$$

Thus  $\Omega$  is gauge invariant.

**Proposition 4.1** (Curvature invariance). *Reference transformations cannot change  $\Omega$ .*

*Proof.*

$$\Omega' = -d(A + d\phi) \quad (24)$$

$$= -dA \quad (25)$$

$$= \Omega, \quad (26)$$

where  $d^2\phi = 0$ .  $\square$

The local components of the invariant residual may be written

$$\Theta = \Theta_t dt + \Theta_x dx + \Theta_y dy + \Theta_z dz. \quad (27)$$

This is not spacetime curvature in the Einstein sense;  $\Omega$  is the curvature of the proposed temporal comparison connection. Relativistic proper time remains determined by the spacetime metric and the observer's worldline [9,10].

For a path  $\gamma$ , define residual accumulation by

$$\mathcal{R}_\gamma = \int_\gamma \Theta. \quad (28)$$

For a closed path,

$$H_\gamma = \oint_\gamma \Theta. \quad (29)$$

Stokes' theorem yields

$$H_\gamma = \int_S \Omega, \quad (30)$$

whenever  $\partial S = \gamma$ . Thus nonzero temporal curvature generates path-dependent residual accumulation, or temporal holonomy.

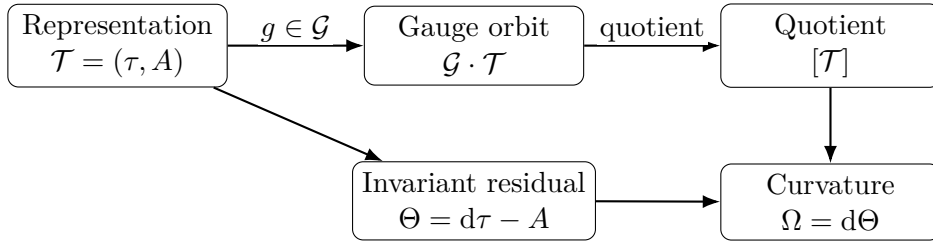


Figure 1: Reference-free CTR: a representation, its gauge orbit, and invariant temporal structure.

## 5 Observational Equivalence and Temporal Fakeness

The earlier fakeness framework distinguishes a changed representation from a changed observable. In logistic regression, affine transformations can preserve the linear predictor after the coefficients are transformed dually, so predictions remain invariant although the feature representation changes [1]. CTR generalizes the same logic to temporal observers.

Let

$$\mathcal{O} : \mathcal{T} \rightarrow \mathcal{Y} \quad (31)$$

be an observer, estimator, classifier, or decision rule. Define

$$\mathcal{T}_1 \sim_{\mathcal{O}} \mathcal{T}_2 \iff \mathcal{O}(\mathcal{T}_1) = \mathcal{O}(\mathcal{T}_2). \quad (32)$$

Its symmetry group is

$$\mathcal{G}_{\mathcal{O}} = \{g : \mathcal{O}(g \cdot \mathcal{T}) = \mathcal{O}(\mathcal{T})\}. \quad (33)$$

**Definition 5.1** (Gauge-minimized temporal fakeness). *For a distance  $d$  on representations, define*

$$F_{\text{gauge}}(\mathcal{T}, \tilde{\mathcal{T}}) = \inf_{g \in \mathcal{G}} d(g \cdot \mathcal{T}, \tilde{\mathcal{T}}). \quad (34)$$

Define the normalized domain average by  $\mathbb{E}_D[\cdot]$ . More explicitly,  $\mathbb{E}_D[\cdot] = \text{Vol}(D)^{-1} \int_D (\cdot) d\mu$ . Let

$$\Delta_\Theta = \tilde{\Theta} - \Theta. \quad (35)$$

The intrinsic distortion is

$$F_{\text{int}} = \mathbb{E}_D[\Delta_\Theta^\mu \Delta_{\Theta\mu}]. \quad (36)$$

A transformation may therefore have substantial representation-level displacement while satisfying  $F_{\text{int}} = 0$ . This is the temporal analogue of gauge-preserving fakeness [1].

Two additional diagnostics follow the same architecture:

$$F_\lambda = \sum_i |\lambda_i - \tilde{\lambda}_i|, \quad (37)$$

$$F_{\text{KL}} = D_{\text{KL}}(P \parallel Q), \quad (38)$$

where the first compares residual covariance spectra and the second compares probability laws of the invariant residual field.

Table 2: Reference-free measures.

| Measure                | Mathematical object | Interpretation                                  |
|------------------------|---------------------|-------------------------------------------------|
| Gauge fakeness         | $F_{\text{gauge}}$  | Representation displacement modulo symmetry     |
| Intrinsic distortion   | $F_{\text{int}}$    | Change in invariant residual structure          |
| Spectral distortion    | $F_\lambda$         | Change in covariance eigenstructure             |
| Information distortion | $F_{\text{KL}}$     | Change in residual distribution                 |
| Holonomy               | $H_\gamma$          | Integrated temporal inconsistency around a loop |

**Theorem 5.1** (Gauge-invariant observers). *If an observer depends on  $\mathcal{T}$  only through  $\Theta$  and gauge-invariant derivatives of  $\Theta$ , then gauge-equivalent representations are observationally equivalent.*

*Proof.* Gauge transformations leave  $\Theta$  unchanged. Any functional of that invariant field and its invariant derivatives therefore has the same value on every representative of a gauge orbit.  $\square$

## 6 Statistics, Econometrics, and Data Science

Reference choice is ubiquitous in statistical modeling. Demeaning, detrending, fixed effects, factor extraction, and residualization all define a subspace regarded as common or nuisance before the remaining structure is analyzed [2,11,12]. RF-CTR makes the choice explicit: a reference is a projection or gauge fixing, while the residual is not automatically noise.

For a panel model,

$$y_{it} = \alpha_i + \lambda_t + u_{it}, \quad (39)$$

the time effect  $\lambda_t$  and unit effects  $\alpha_i$  are model-selected common components. The analogy with CTR is structural rather than literal: in both cases, what remains after projection may contain the economically relevant organization [11].

For stochastic temporal fields, define the covariance kernel

$$K_{\mu\nu}(p, q) = \mathbb{E}[\Theta_\mu(p)\Theta_\nu(q)]. \quad (40)$$

This retains cross-location information lost by a single standard deviation. Gaussian-process, state-space, time-series, and covariance-operator methods provide natural estimation layers [3,4,13].

## 7 Physics and the Natural Sciences

Relativity distinguishes proper time along worldlines from coordinate descriptions of space-time, and synchronization depends on the chosen observational construction [9,10]. RF-CTR is compatible with that distinction but is not a replacement for relativistic dynamics.

In biology, coupled circadian oscillators and synchronization provide a natural empirical domain for residual fields: local clock phases, common entrainment, and phase differences can be analyzed without assuming that all observed timing heterogeneity is independent noise [14,15]. In chemical kinetics, reaction progress and relaxation modes similarly show why deviations from a fitted temporal background may carry mechanism rather than mere error [16]. None of these analogies implies that biological or chemical time is itself a gauge field.

Psychological work shows that experienced duration varies with attention, emotion, and context [17,18]. This motivates a distinction between a physical clock field and observer-specific temporal representations. The psychiatric use of CTR should be limited to descriptive modeling of reported temporal experience; the framework is not a diagnostic instrument.

## 8 Economics, Finance, Welfare, and Political Economy

In finance, asynchronous observations and latent common factors complicate a single market clock. Returns may be represented as

$$r_{it} = \alpha_i + \beta_i^\top f_t + \varepsilon_{it}, \quad (41)$$

where residual dependence can itself be economically informative [19,20]. RF-CTR asks which temporal conclusions survive alternative synchronization and reference conventions.

Welfare economics adds an aggregation issue. For welfare weights  $w_i$  and intertemporal utilities  $u_i$ , one may write

$$W = \sum_i w_i \int u_i(t) dt. \quad (42)$$

A common temporal reference can simplify aggregation while obscuring heterogeneous delays or timing exposures. A residual-field representation postpones aggregation until after temporal heterogeneity has been measured [21,22].

Political science and international relations often infer latent conditions from noisy or strategically reported indicators. The fakeness framework establishes that an observer can be invariant to substantial representational changes [1]; strategic signaling theory emphasizes that information is filtered through beliefs, institutions, and strategic interaction [23,24]. RF-CTR provides a mathematical vocabulary for asking whether an apparent temporal change is a genuine invariant change or a change of reference and representation.

For warfare economics and military analysis, the framework is intentionally abstract: it treats deception as manipulation of an observer's representation and invariant information, rather than supplying operational guidance [25,26].

## 9 Philosophy, Religion, and Epistemology

The epistemological claim is modest: an estimate of common time is not identical to the observations from which it is computed. The distinction resembles the broader philosophical difference between appearance, representation, and what remains invariant under a change of description [27,28]. Augustine’s reflections on temporal experience and modern philosophy of time both illustrate that operational clock time and lived or metaphysical conceptions of time need not be identical [29,30].

Religious texts likewise contain temporal ontologies that need not coincide with laboratory clock conventions. The Bhagavad Gita, for example, places human action within a broader conception of temporal order [31]. Such material is interpretive context, not scientific evidence for the proposed gauge calculus.

## 10 AI, Machine Learning, and Algorithmic Construction

A learning system observes a representation through an operator  $\mathcal{O}$ . The central computational question is therefore not simply whether the input changed, but whether the change survives the observer’s symmetry group. This generalizes the representation-equivalence result of [1] and is compatible with the broader treatment of invariance and representation learning in machine learning [5,32].

A minimal implementation is:

```
INPUT: temporal observations tau on D
        comparison one-form A
        admissible group G
        observer O
```

1. Form  $T = (\tau, A)$ .
  2. Generate equivalent representatives  $g(T)$ ,  $g$  in  $G$ .
  3. Compute  $\Theta = d\tau - A$ .
  4. Compute  $\Omega = d\Theta$ .
  5. Estimate covariance, spectra, and structure functions.
  6. Compute gauge-minimized and intrinsic distortions.
  7. Evaluate observational equivalence under  $O$ .
  8. Report invariants, uncertainty, and sensitivity to  $G$ .
- OUTPUT: gauge class, invariant temporal structure, diagnostics

For discretized data, finite-difference or finite-element exterior derivatives, sparse linear algebra, state-space models, Gaussian processes, and neural sequence models can provide implementation layers [3,4,32]. The symmetry group and observer should be declared before inference to prevent post hoc gauge selection.

## 11 Empirical Program and Testable Claims

The first empirical stage should be synthetic. Choose  $\tau$ , choose a nontrivial  $\phi$ , construct

$$\tilde{\tau} = \tau + \phi, \tag{43}$$

$$\tilde{A} = A + d\phi, \tag{44}$$

and verify numerically that

$$\tilde{\Theta} = \Theta. \tag{45}$$

Then introduce deliberately non-gauge perturbations and test whether  $F_{\text{intrinsic}}$ ,  $F_{\lambda}$ , and  $F_{\text{KL}}$  detect them.

A useful hypothesis pair is

$$H_0 : \text{the observed change is gauge-equivalent,} \quad (46)$$

$$H_1 : \text{the observed change alters invariant temporal structure.} \quad (47)$$

Suitable real-data applications include distributed sensor networks, financial tick data, biological oscillator records, computing logs, and repeated-measures panels. Robustness should be assessed over alternative measures, discretizations, observers, and admissible gauge groups. Bootstrap, likelihood, Bayesian, or permutation methods can be chosen according to the data-generating process [2–4,11,13].

## 12 Scope and Limitations

Three cautions are fundamental. First,  $A$  is a modeling device for temporal comparison;  $\Omega$  is not automatically a new physical interaction. Second, the gauge group is part of the model. If it is too large, substantively different structures can be identified; if it is too small, representational artifacts may survive. Third, reference-free does not mean error-free: measurement error, omitted variables, selection, dependence, and misspecification remain.

The strongest claim is therefore methodological rather than metaphysical:

reference dependence can be separated from invariant temporal structure.

(48)

## 13 Conclusion

CTR begins with the observation that common time is calculated from local clock readings. Its reference-dependent form decomposes a temporal field into a projection and a residual. Its reference-free form treats admissible changes of temporal reference as gauge transformations and identifies the invariant field

$$\Theta = d\tau - A. \quad (49)$$

Its curvature is

$$\Omega = d\Theta, \quad (50)$$

and its finite relational content is captured by line integrals and holonomy.

The framework also imports the central lesson of gauge-invariant fakeness: a representation may change substantially while an observer remains exactly invariant [1]. Temporal gauge fakeness and intrinsic distortion therefore separate representational deformation from genuine changes in invariant temporal structure.

The resulting hierarchy is

$$\text{observations} \rightarrow \text{reference representation} \rightarrow \text{gauge orbit} \quad (51)$$

$$\rightarrow \text{invariant residual field} \rightarrow \text{curvature and holonomy.} \quad (52)$$

The conceptual inversion can be stated in one sentence:

A reference may be conventional even when the residual structure is not.

(53)

## References

- [1] S. Ghosh and OpenAI, “Gauge-Invariant Measures of Fakeness under Logistic Regression: A Mathematical, Statistical, Economic, and Geopolitical Theory of Invisible Transformations,” 2026.
- [2] J. M. Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, 2nd ed., MIT Press, 2010.
- [3] J. D. Hamilton, *Time Series Analysis*, Princeton University Press, 1994.
- [4] C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*, MIT Press, 2006.
- [5] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009.
- [6] H. Weyl, “Gravitation und Elektrizität,” *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin*, 465–480, 1918.
- [7] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed., Institute of Physics Publishing, 2003.
- [8] R. M. Wald, *General Relativity*, University of Chicago Press, 1984.
- [9] A. Einstein, “Zur Elektrodynamik bewegter Körper,” *Annalen der Physik*, 17, 891–921, 1905.
- [10] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation*, W. H. Freeman, 1973.
- [11] J. D. Angrist and J.-S. Pischke, *Mostly Harmless Econometrics*, Princeton University Press, 2009.
- [12] J. Y. Campbell, A. W. Lo, and A. C. MacKinlay, *The Econometrics of Financial Markets*, Princeton University Press, 1997.
- [13] C. A. Czeisler et al., “Stability, Precision, and Near-24-Hour Period of the Human Circadian Pacemaker,” *Science*, 284(5423), 2177–2181, 1999.
- [14] A. T. Winfree, *The Geometry of Biological Time*, 2nd ed., Springer, 2001.
- [15] K. J. Laidler, *Chemical Kinetics*, 3rd ed., Harper & Row, 1987.
- [16] M. Wittmann, “The Inner Experience of Time,” *Philosophical Transactions of the Royal Society B*, 368, 20120408, 2013.
- [17] S. Droit-Volet and W. H. Meck, “How Emotions Colour Our Perception of Time,” *Trends in Cognitive Sciences*, 11(12), 504–513, 2007.
- [18] E. F. Fama, “Efficient Capital Markets: A Review of Theory and Empirical Work,” *Journal of Finance*, 25(2), 383–417, 1970.
- [19] A. Sen, *Development as Freedom*, Oxford University Press, 1999.
- [20] K. J. Arrow, *Social Choice and Individual Values*, Wiley, 1951.
- [21] T. C. Schelling, *The Strategy of Conflict*, Harvard University Press, 1960.

- [22] R. Jervis, *Perception and Misperception in International Politics*, Princeton University Press, 1976.
- [23] Sun Tzu, *The Art of War*, various translations.
- [24] C. von Clausewitz, *On War*, 1832.
- [25] Augustine, *Confessions*, Book XI, various translations.
- [26] D. H. Mellor, *Real Time II*, Routledge, 1998.
- [27] E. Easwaran, trans., *The Bhagavad Gita*, 2nd ed., Nilgiri Press, 2007.
- [28] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.

## Glossary

### Temporal field

A map  $\tau : M \rightarrow \mathbb{R}$  assigning temporal observations to spacetime events.

### Reference projection

A rule such as  $P_D$  selecting a common component.

### Temporal residual

The component left after the chosen reference is removed.

### Gauge transformation

A representation change preserving designated observables.

### Gauge orbit

The set  $\mathcal{G} \cdot \mathcal{T}$  of equivalent representations.

### Quotient space

A space obtained by identifying representations on the same orbit.

### Temporal comparison connection

The one-form  $A$  defining local temporal comparison.

### Invariant residual

The one-form  $\Theta = d\tau - A$ .

### Temporal curvature

The two-form  $\Omega = d\Theta$ .

### Temporal holonomy

The loop integral  $H_\gamma = \oint_\gamma \Theta$ .

### Observational equivalence

Equality of outputs under an observer  $\mathcal{O}$ .

### Temporal gauge fakeness

Representation displacement after quotienting by admissible symmetries.

### Intrinsic temporal distortion

Change in invariant temporal structure.

## The End